

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented a **Python/FastAPI backend** for integrating new AI supports, focusing on **prompt engineering** and **API orchestration** for curriculum-aligned answers.
- Developed a **Retrieval-Augmented Generation (RAG) pipeline**, optimizing data retrieval and LLM integration to ensure accurate, context-aware responses for educational content.
- Owned the **UI localisation** for the existing teaching and learning platform, wiring **React** components to backend logic to deliver a seamless, accessible user experience.
- Architected technical groundwork for expansion into the primary school market, adapting the white-label platform infrastructure and data models for future scalability.
- Collaborated with AI engineers at a University Partner and external platform builders to refine early-stage AI Proof of Concept into market-ready features.
- Containerized AI backend services using **Docker** and deployed them to a Kubernetes cluster, managing configurations and resource allocation for stable operation.
- Integrated AI-generated content with existing platform features, implementing **error handling** and **data validation** to maintain system integrity and user trust.

SKILLS

AI/ML – Python, FastAPI, Prompt Engineering, API Orchestration, RAG, LLM, scikit-learn.

Backend Development – Java, Spring Boot, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Oracle.

Frontend Development – React, Angular, HTML, CSS, JavaScript, TypeScript.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git.

Data & Analytics – Power BI, Tableau, DAX, Power Query, pandas, NumPy, R, Data Modelling.

PROJECTS

AI-Powered Tutoring Backend Integration

- Developed a **Python/FastAPI** backend service to orchestrate calls between a user-facing application and an external LLM provider for AI-assisted tutoring.
- Implemented a **Retrieval-Augmented Generation (RAG)** system, integrating a vector database (e.g., Pinecone, Weaviate concepts) to fetch relevant educational materials for context-aware AI responses, reducing irrelevant answers by **~40%**.
- Designed and built RESTful APIs for seamless integration with the frontend application, ensuring efficient data exchange and error handling for a consistent user experience.

Platform Localisation and UI Enhancement

- Enhanced an existing **React**-based teaching platform by implementing new UI components and features for improved localisation and accessibility, supporting **bilingual content delivery**.
- Connected frontend components to backend services, ensuring accurate rendering of localised content and seamless user interaction across different language settings.
- Collaborated with UI/UX designers to translate mockups into functional interface elements, contributing to a more engaging and user-friendly learning environment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)